CLAUDE PROJECT UPDATE

MedMasters Course Architecture System™ (MCAS)

Project Overview

The MedMasters Course Architecture System™ (MCAS) is a faculty-centered instructional design and course development framework designed to help educators build high-quality, accessible, compliant, engaging, and scalable courses.

MCAS is not primarily a content generation system.

MCAS is a course architecture system that organizes instructional design, compliance requirements, course assets, assessment planning, student engagement, and AI-assisted content development into a single repeatable workflow.

The goal is to create a complete Course Operating System that can be used by faculty, instructional designers, departments, and institutions.

Core Philosophy

Most AI tools generate content.

MedMasters architects learning experiences.

The system should prioritize:

1. Learning outcomes before content.
2. Student experience before technology.
3. Course architecture before AI generation.
4. Compliance before deployment.
5. Reusability and scalability.
6. Evidence-based instructional design.
7. Reduced faculty workload through organization and automation.

Key Message:

"Great courses aren't generated. They're architected."

Current System Architecture

The current approved workflow is:

DISCOVER ↓ DESIGN ↓ ASSET CAPTURE ↓ SCHEDULE & PACING ↓ FRAME ↓ BUILD ↓ ASSESS ↓ ENGAGE ↓ COMPLIANCE ↓ LAUNCH ↓ SCALE

Phase 1: Discover

Purpose: Define the course before building.

Outputs:

- Course Vision
- Student Population
- Delivery Method
- Course Scope
- Learning Outcomes
- Weekly Rhythm
- Success Metrics

Deliverable:

Course Design Brief

Phase 2: Design

Purpose: Create the instructional identity.

Outputs:

- Faculty Brand Sheet
- Teaching Philosophy
- Faculty Voice Guide
- Visual Style Guide
- Accessibility Preferences
- AI Preferences

Deliverable:

Master Faculty Brand Sheet

The Faculty Brand Sheet serves as the instructional DNA for all future AI-generated content.

Phase 3: Asset Capture

Purpose: Collect and organize existing faculty resources.

Examples:

- PowerPoints
- Lecture Notes
- PDFs
- Study Guides
- Images
- YouTube Videos
- Recorded Lectures
- Lab Activities
- Publisher Resources
- Existing LMS Content

Deliverable:

Asset Inventory

Important Principle:

Faculty already have valuable materials.

The system should organize and leverage existing resources before generating new content.

Phase 4: Schedule & Pacing

Purpose: Distribute learning across the term.

Outputs:

- Semester Map
- Weekly Calendar
- Assessment Calendar
- RSI Calendar
- Student Workload Analysis
- Faculty Workload Analysis

Deliverable:

Master Schedule

The system should identify workload bottlenecks and provide recommendations.

Phase 5: Frame

Purpose: Create the course structure.

Outputs:

- Module Blueprint
- Weekly Templates
- Navigation Structure
- Alignment Map
- Assessment Map

Deliverable:

Master Course Blueprint

Phase 6: Build

Purpose: Generate course materials.

Possible Outputs:

- Welcome Pages
- Study Guides
- Lecture Scripts
- Notes
- Worksheets
- Assignments
- Discussions
- Labs
- Resource Pages

Deliverable:

Course Content Package

Phase 7: Assess

Purpose: Measure learning.

Possible Outputs:

- Quizzes
- Exams
- Rubrics
- Projects
- Practical Assessments
- Clinical Assessments

Deliverable:

Assessment Package

Phase 8: Engage

Purpose: Create active learning experiences.

Potential Outputs:

- Team-Based Learning Activities
- Clinical Cases
- Simulations
- Draw-to-Know-It Activities
- Map & Master Activities
- Reflection Activities
- Application Activities

Deliverable:

Engagement Toolkit

Special MedMasters Differentiators:

- TBL Engine
- Clinical Case Engine
- Sim Patient Engine
- AI-Resistant Assignment Builder
- Oral Assessment Builder

Phase 9: Compliance

This phase is a major differentiator and should be treated as a primary system component.

Accessibility Compliance

Standard:

WCAG 2.2 Level AA

Review Areas:

- Heading Structure
- Alt Text
- Color Contrast
- Accessible Tables
- Captioning
- Transcripts
- Accessible Documents
- Keyboard Navigation

Deliverable:

Accessibility Compliance Report

The system should generate evidence showing how the course complies with accessibility requirements.

RSI Compliance

Standard:

Regular and Substantive Interaction (RSI)

Review Areas:

- Instructor Presence
- Direct Instruction
- Guided Facilitation
- Feedback Schedule
- Communication Schedule
- Office Hours
- Student Support

Deliverable:

RSI Compliance Report

The system should generate an RSI Evidence Matrix showing where and how RSI occurs throughout the course.

Example:

Week Instructor Activity Student Activity Interaction Type

The report should be suitable for institutional review.

Academic Integrity

Review Areas:

- AI Policies
- Authentic Assessments
- Academic Integrity Statements
- Assessment Design
- Integrity Safeguards

Deliverable:

Academic Integrity Review

Quality Assurance

Review Areas:

- Alignment
- Navigation
- Student Experience
- Consistency
- Outcome Coverage

Deliverable:

Quality Assurance Report

Phase 10: Launch

Purpose: Prepare the course for deployment.

Outputs:

- Final Audit
- Accessibility Verification
- RSI Verification
- Readiness Review
- Launch Checklist

Deliverable:

Course Launch Package

Phase 11: Scale

Purpose: Expand and improve.

Outputs:

- Companion Courses
- Program Alignment
- Hybrid Versions
- Online Versions
- Faculty Collaboration Resources
- Shared Asset Libraries

Deliverable:

Growth Plan

Master Workbook Strategy

The system should use a single workbook as the primary intake mechanism.

Current Workbook Structure:

1. Course Information
2. Faculty Brand Sheet
3. Learning Outcomes
4. Existing Assets
5. Schedule & Pacing
6. Module Builder
7. Assessments
8. Engagement Activities

9. Accessibility Review
10. RSI Review
11. Quality Assurance
12. Launch Checklist

The workbook serves as the single source of truth.

All AI-generated outputs should originate from workbook data.

Planned Enhancements for Workbook Version 2

Future workbook features:

- MedMasters Branding
- Dashboard
- Progress Tracker
- Automated Calculations
- Readiness Score
- WCAG Scoring
- RSI Scoring
- Compliance Dashboard
- Example Entries
- Embedded Instructions
- Faculty-Friendly Navigation

Final Certification Package

The long-term vision includes automatic generation of:

- Course Blueprint Report
- Assessment Architecture Report
- Accessibility Compliance Report
- RSI Compliance Report
- Academic Integrity Report
- Quality Assurance Report
- Launch Checklist
- Course Readiness Certification

These reports should be suitable for faculty, instructional designers, department chairs, deans, and accreditation reviewers.

Long-Term Vision

The long-term goal is to develop a MedMasters Course Operating System™ that helps faculty:

- Organize existing resources
- Build aligned courses
- Meet accessibility requirements
- Demonstrate RSI compliance
- Improve student engagement
- Reduce development time
- Produce institutional-quality documentation

The system should function as both a course design framework and a course quality assurance system.

Every recommendation, tool, template, workflow, and future development should align with this vision.